

Problem#1: Solving Lakota Equations (15 marks)

Solution:

The Lakota numbers used in this problem are:

Wanji = 1; *Nunpa* = 2; *Yamni* = 3; *Topa* = 4; *Zaptan* = 5;
Šakpe = 6; *Šaglohan* = 8; *Nepcunka* = 9; *Wikcemna* = 10

Some other rules for specifying numbers in this language are:

- *opawing'e* = 100
- *Ake* $x = 10 + x$
- *wikcemna* $x = 10x$
- *opawing'e sam* $x = 100 + x$

Assignment 1:

- a) *wikcemna + nunpa* = *ake nunpa* (multiple answers possible e.g. *Nepcunka + Yamni*)
- b) *ake wanji + wanji* = *nunpa + wikcemna*
- c) *wikcemna yamni* = *zaptan* × *Šakpe*

Assignment 2:

- d) *ake topa* = 14
- e) *opawing'e sam yamni* = 103

Assignment 3:

- f) 13 = *ake yamni*
 - g) 20 = *wikcemna nunpa*
-